

Supplementary materials for “Ultrafast diffraction from molecular symmetry in photoelectron distribution”

Sajad Azizi,^{1,*} Mohamed El-Amine Madjet,² Zheng Li,^{3,4,5} Jan M. Rost,⁶ and Himadri S. Chakraborty^{2,†}

¹Leibniz Supercomputing Centre, Boltzmannstraße 1, 85748 Garching bei München, Germany

²Department of Natural Sciences, Dean L. Hubbard Center for Innovation,
Northwest Missouri State University, Maryville, Missouri 64468, USA

³State Key Laboratory for Mesoscopic Physics and Collaborative Innovation Center of Quantum Matter,
School of Physics, Peking University, Beijing 100871, China

⁴Collaborative Innovation Center of Extreme Optics, Shanxi University, Taiyuan, Shanxi 030006, China

⁵Peking University Yangtze Delta Institute of Optoelectronics, Nantong, Jiangsu 226010, China

⁶Max Planck Institute for the Physics of Complex Systems, Nöthnitzer Strasse 38, D-01187, Dresden, Germany

(Dated: May 18, 2026)

I. MOLECULAR SYSTEM

We consider the single-photon ionization of a molecule within the fixed-nuclei approximation. All coordinates are referred to a body-fixed frame whose origin is placed at the center of mass, and the molecular geometry is kept frozen at its equilibrium nuclear configuration. The electronic structure of the residual ion is described at the Hartree–Fock level: the occupied orbitals Φ^i ($i \in \text{occ}$) are not re-optimized in the presence of the continuum electron (static-exchange approximation). The outgoing photoelectron therefore moves in the combined static (Coulomb + nuclear) and nonlocal exchange field of the frozen ionic core.

The full Hamiltonian, including the coupling to the radiation field in the length gauge, reads

$$H = \frac{p^2}{2} + V(\mathbf{r}) + \mathbf{r} \cdot \mathbf{F}, \quad (\text{S1})$$

where V denotes the effective one-electron potential of the static-exchange approximation (comprising both local and nonlocal parts, as specified below) and $\mathbf{r} \cdot \mathbf{F}$ is the dipole interaction with the electric field $\mathbf{F}$ and $|\mathbf{F}| = 1$. Because the photon interaction is treated in first-order perturbation theory, the photoionization amplitude reduces to the dipole matrix element $d_{\ell m} = \langle \Psi_{\ell m}(E) | \mathbf{r} \cdot \mathbf{F} | \Psi_i \rangle$ [Eq. (S23)], and the central task of this section is to obtain the field-free continuum states $\Psi_{\ell m}$. These satisfy the integro-differential equation

$$\left(-\frac{1}{2} \nabla^2 + V(\mathbf{r}) \right) \Psi(\mathbf{r}) = E \Psi(\mathbf{r}), \quad (\text{S2})$$

where $E > 0$ is the continuum energy. Here V is not merely a local function; its action on the continuum orbital Ψ is the sum of a local static potential $V_{\text{st}}(\mathbf{r})$ (electron–nuclear attraction plus classical Coulomb repulsion with the occupied

orbitals), polarization potential, and a nonlocal exchange operator V_{exc} ,

$$V(\mathbf{r}) \Psi(\mathbf{r}) = (V_{\text{pol}}(\mathbf{r}) + V_{\text{st}}(\mathbf{r})) \Psi(\mathbf{r}) - (V_{\text{exc}} \Psi)(\mathbf{r}). \quad (\text{S3})$$

Substituting Eq. (S3) into Eq. (S2) and rearranging yields

$$\left(\nabla^2 + 2E - 2(V_{\text{pol}}(\mathbf{r}) + V_{\text{st}}(\mathbf{r})) \right) \Psi(\mathbf{r}) = -2(V_{\text{exc}} \Psi)(\mathbf{r}). \quad (\text{S4})$$

The nonlocal exchange term acts on Ψ through all occupied Hartree–Fock orbitals,

$$(V_{\text{exc}} \Psi)(\mathbf{r}) = \sum_{i \in \text{occ}} \Phi^i(\mathbf{r}) F^i(\mathbf{r}), \quad (\text{S5})$$

where the auxiliary functions F^i are defined by the Coulomb integrals

$$F^i(\mathbf{r}) = \int d^3 r' \frac{\Phi^{i*}(\mathbf{r}') \Psi(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \quad (\text{S6})$$

These satisfy Poisson’s equation,

$$\nabla^2 F^i(\mathbf{r}) = -4\pi \Phi^{i*}(\mathbf{r}) \Psi(\mathbf{r}). \quad (\text{S7})$$

The long-range polarization interaction is modeled as

$$V_{\text{pol}}(\mathbf{r}) = -\frac{1}{2r^4} f_c(r, r_c) [\alpha_0 + \alpha_2 P_2(\cos \theta)], \quad (\text{S8})$$

where

$$f_c(r, r_c) = 1 - \exp\left[-\left(\frac{r}{r_c}\right)^n\right] \quad (\text{S9})$$

is a short-range cutoff function introduced to regularize the polarization potential at small r , and $P_2(\cos \theta) = \frac{1}{2}(3\cos^2 \theta - 1)$ is the second Legendre polynomial. In the present work, we use $\alpha_0 = 78.41841$ a.u., $r_c = 4.3 a_0$, and $n = 6$.

Because perfluorocubane (C_8F_8) belongs to the cubic point group O_h , its dipole polarizability tensor is isotropic. Therefore, the rank-2 anisotropic contribution vanishes, and we set $\alpha_2 = 0$.

* azizi.dajas@gmail.com

† himadri@nwmissouri.edu

All quantities are expanded about the chosen origin in a single-center partial-wave basis of spherical harmonics $Y_{\ell,m}$ with $\langle Y_{\ell,m} | Y_{\ell',m'} \rangle = \delta_{\ell\ell'} \delta_{mm'}$. We adopt the composite angular-momentum indices $\mu \equiv (\ell_\mu, m_\mu)$, $\sigma \equiv (\ell_\sigma, m_\sigma)$, and $\lambda \equiv (\ell_\lambda, m_\lambda)$ to label partial waves, and write

$$\Psi(r) = \sum_{\mu} \frac{\psi_{\mu}(r)}{r} Y_{\mu}(\hat{r}), \quad (\text{S10a})$$

$$F^i(r) = \sum_{\sigma} \frac{f_{\sigma}^i(r)}{r} Y_{\sigma}(\hat{r}), \quad (\text{S10b})$$

$$\Phi^i(r) = \sum_{\lambda} \frac{\chi_{\lambda}^i(r)}{r} Y_{\lambda}(\hat{r}). \quad (\text{S10c})$$

Inserting Eqs. (S10) into the Poisson equation (S7) and projecting onto Y_{σ}^* gives the radial equation

$$\begin{aligned} f_{\sigma}^{i''}(r) - \frac{\ell_{\sigma}(\ell_{\sigma} + 1)}{r^2} f_{\sigma}^i(r) \\ = -4\pi \sum_{\lambda, \mu} C_{\mu, \lambda, \sigma} \frac{\chi_{\lambda}^{i*}(r) \psi_{\mu}(r)}{r}, \end{aligned} \quad (\text{S11})$$

where $C_{\mu, \lambda, \sigma} \equiv \int Y_{\mu} Y_{\lambda}^* Y_{\sigma}^* d\Omega$ are Gaunt coefficients. The boundary conditions for the Poisson radial functions are

$$f_{\sigma}^i(r) \sim r^{\ell_{\sigma}+1} \quad \text{as } r \rightarrow 0, \quad (\text{S12a})$$

$$f_{\sigma}^i(r) \sim r^{-\ell_{\sigma}} \quad \text{as } r \rightarrow \infty. \quad (\text{S12b})$$

Likewise, projecting Eq. (S4) onto Y_{μ}^* yields the radial Schrödinger equation

$$\begin{aligned} \sum_{\mu'} [\psi_{\mu'}''(r) \delta_{\mu\mu'} + 2(E \delta_{\mu\mu'} - V_{\mu\mu'}(r)) \psi_{\mu'}(r)] \\ = -2 \sum_{i \in \text{occ}} \sum_{\sigma} Q_{\mu\sigma}^i(r) f_{\sigma}^i(r), \end{aligned} \quad (\text{S13})$$

where $V_{\mu\mu'}(r)$ are the matrix elements of the static potential in the partial-wave basis plus the centrifugal term and polarization. The coupling coefficients on the right-hand side are

$$Q_{\mu\sigma}^i(r) = \sum_{\lambda} G_{\mu\lambda\sigma} \frac{\chi_{\lambda}^i(r)}{r}, \quad (\text{S14})$$

with $G_{\mu\lambda\sigma} \equiv \int Y_{\mu}^* Y_{\lambda} Y_{\sigma} d\Omega$ a Gaunt coefficient that differs from $C_{\mu, \lambda, \sigma}$ in the placement of the complex conjugation.

Equations (S11) and (S13) are combined into a single coupled system. Defining the solution vector $\mathbf{Y} = (\psi, \mathbf{f})^T$ and a coupling matrix $\mathbf{U}$, the system takes the form

$$\mathbf{Y}''(r) - \mathbf{U}(r) \mathbf{Y}(r) = 0, \quad (\text{S15})$$

or, equivalently,

$$\begin{pmatrix} \psi'' \\ \mathbf{f}'' \end{pmatrix} = \begin{pmatrix} U_{\psi\psi} & U_{\psi f} \\ U_{f\psi} & U_{ff} \end{pmatrix} \begin{pmatrix} \psi \\ \mathbf{f} \end{pmatrix}. \quad (\text{S16})$$

The off-diagonal blocks satisfy $U_{\psi f} = U_{f\psi}^{\dagger}$, and the block U_{ff} is a diagonal block. The expansion retains $N_{\text{ch}} = (\ell_{\text{max}} + 1)^2$ partial-wave channels for the continuum orbital, and the coupled system is integrated outward from the origin using the renormalized Numerov method [2] (see Sec. V for implementation parameters).

To construct the scattering matrix one needs N_{ch} linearly independent solutions, which we label by a second composite index μ' running over the same set as μ . The radial amplitude of the μ' -th solution in channel μ is $\psi_{\mu\mu'}(r)$ (and analogously $f_{\sigma\mu'}^i(r)$). Each solution is regular at the origin, $\psi_{\mu\mu'}(r) \rightarrow r^{\ell_{\mu}+1} \delta_{\mu\mu'}$ as $r \rightarrow 0$. In the asymptotic region, where the potential vanishes, the numerical solutions are matched to the Riccati-Bessel functions $\hat{j}_{\ell}(x) = x j_{\ell}(x)$ and $\hat{n}_{\ell}(x) = x n_{\ell}(x)$ (with j_{ℓ} , n_{ℓ} the spherical Bessel and Neumann functions) to extract the reactance matrix $\mathbf{K}$, from which the scattering matrix follows as $\mathbf{S} = (\mathbf{I} + i\mathbf{K})(\mathbf{I} - i\mathbf{K})^{-1}$. The resulting S-matrix solutions satisfy

$$\psi_{\mu\mu'}^S(r) \sim \hat{j}_{\ell_{\mu}}(kr) \delta_{\mu\mu'} \quad \text{as } r \rightarrow 0, \quad (\text{S17a})$$

and

$$\begin{aligned} \psi_{\mu\mu'}^S(r) \sim \frac{1}{2} \hat{h}_{\ell_{\mu}}^+(kr) \delta_{\mu\mu'} \\ + \frac{1}{2} \hat{h}_{\ell_{\mu}}^-(kr) S_{\mu\mu'} \quad \text{as } r \rightarrow \infty, \end{aligned} \quad (\text{S17b})$$

where $k = \sqrt{2E}$ is the photoelectron wavenumber, $\hat{h}_{\ell}^{\pm} = \hat{j}_{\ell} \pm i \hat{n}_{\ell}$ are the Riccati-Hankel functions ($\hat{h}^+$ incoming, $\hat{h}^-$ outgoing), and both indices of $S_{\mu\mu'}$ are composite angular-momentum labels of the type (ℓ, m) defined above.

The energy-normalized partial-wave continuum states $\Psi_{\ell'm'}(E, r)$ are reconstructed from the S-matrix solutions $\psi_{\mu\mu'}^S$ by resummation over all channels μ with a normalization prefactor of $\sqrt{2/(\pi k)}$, which follows from the unitarity of $\mathbf{S}$ and ensures the orthonormality condition $\langle \Psi_{\ell m}(E) | \Psi_{\ell' m'}(E') \rangle = \delta_{\ell\ell'} \delta_{mm'} \delta(E - E')$. [we combine this prefactor to prefactor in the final wavefunction, eq. S19](#) These are the states that enter the partial-wave expansion of the final-state wavefunction [Eq. (S19)] and the dipole matrix elements $d_{\ell m}$ that determine the cross section and photoionization time delay (Sec. II).

To isolate the role of molecular geometry in photoelectron diffraction, we also employ a local model potential with cubic symmetry,

$$V(r) = \begin{cases} -V_0, & |x| < \frac{L}{2}, |y| < \frac{L}{2}, |z| < \frac{L}{2}, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{S18})$$

which represents the potential well experienced by the delocalized HOMO electron of C_8F_8^- (or the HOMO-to-LUMO excited electron of the neutral molecule). With the parameters $V_0 = 1.65$ a.u. and $L = 1.7$ a.u., the potential supports a ground state at $E_0 = -2.81$ eV. In this model the exchange operator vanishes and the coupled system reduces to the radial Schrödinger equation (S13) alone, with $V_{\mu\mu'}(r)$ given

by the angular matrix elements of Eq. (S18). The cubic geometry imprints a characteristic diffraction pattern on the photoionization cross section and time delay, which can be tracked analytically as the shape is continuously deformed (Sec. III).

II. INTERACTION DYNAMICS

The energy-normalized partial-wave continuum states $\Psi_{\ell m}(E, \mathbf{r})$ obtained in Sec. I are now used to construct the photoionization observables. The final state of the system, comprising the residual ion and the photoelectron with asymptotic momentum $\mathbf{k} = k\hat{\mathbf{k}}$ ($k = \sqrt{2E}$), is expanded in partial waves as

$$\Psi_f^{(-)}(\mathbf{k}, \mathbf{r}) = 4\pi \sum_{\ell m} \frac{i^\ell}{k} \Psi_{\ell m}(E, \mathbf{r}) Y_{\ell m}^*(\hat{\mathbf{k}}), \quad (\text{S19})$$

where $\hat{\mathbf{k}} \equiv (\vartheta, \varphi)$ is the emission direction. The superscript $(-)$ indicates outgoing-wave boundary conditions, so that the state describes a detected electron propagating along $\hat{\mathbf{k}}$.

After the laser pulse, the system is represented by the wavepacket

$$\Phi(\mathbf{r}, t) = \int d\mathbf{k} \phi(\mathbf{k}) \Psi_f^{(-)}(\mathbf{k}, \mathbf{r}) \exp(-i\frac{k^2}{2}t), \quad (\text{S20})$$

where $\phi(\mathbf{k})$ is the ionization probability amplitude. At asymptotically large times the integral is dominated by the stationary-phase point, and the electron with momentum $\mathbf{k} = k\hat{\mathbf{n}}$ arrives at the detector position $\mathbf{r} = r\hat{\mathbf{n}}$ at the time

$$t = \frac{r}{k} + \frac{1}{k} \frac{\partial \arg \phi(\mathbf{k})}{\partial k}. \quad (\text{S21})$$

The first term is the free-flight time from the origin to $\mathbf{r}$; the second is the photoionization time delay,

$$\tau(\mathbf{k}) = \frac{\partial \arg \phi(\mathbf{k})}{\partial E} = \text{Im} \left[\frac{1}{\mathcal{D}^{(-)}(\mathbf{k})} \frac{d\mathcal{D}^{(-)}(\mathbf{k})}{dE} \right], \quad (\text{S22})$$

where the second equality uses $\partial_E \arg[\cdot] = \text{Im}[\cdot^{-1} \partial_E(\cdot)]$ and avoids numerical phase unwrapping.

In the perturbative regime the ionization phase is entirely determined by the dipole transition amplitude. Using the length-gauge form of the interaction $\mathbf{r} \cdot \mathbf{F}$ [Eq. (S1)], this amplitude is expanded in partial waves as

$$\mathcal{D}^{(-)}(\mathbf{k}) = 4\pi \sum_{\ell m} \frac{i^{-\ell}}{k} d_{\ell m}(E) Y_{\ell m}(\hat{\mathbf{k}}), \quad (\text{S23})$$

with the partial dipole matrix elements

$$d_{\ell m}(E) = \langle \Psi_{\ell m}(E, \mathbf{r}) | \mathbf{r} \cdot \mathbf{F} | \Psi_i \rangle, \quad (\text{S24})$$

where Ψ_i is the initial bound state. The angle-resolved cross section is

$$\sigma(\mathbf{k}) = \frac{4\pi^2\omega}{c} |\mathcal{D}^{(-)}(\mathbf{k})|^2. \quad (\text{S25})$$

The electric-field orientation is fixed in the laboratory frame (LF), whereas the position operator $\mathbf{r}$ is naturally expressed in the molecular frame (MF). The two frames are related by the Euler angles $\hat{R} \equiv (\alpha, \beta, \gamma)$, which rotate the LF into coincidence with the MF (Fig. 1 of the main text). The spherical components of the position operator in the MF are $r_\nu = \sqrt{4\pi/3} r Y_1^\nu(\hat{\mathbf{r}})$ with $\hat{\mathbf{r}} \equiv (\theta, \phi)$. For a linearly polarized pulse with $|\mathbf{F}| = 1$, the dipole operator reads

$$\hat{\mu} \equiv \mathbf{r} \cdot \mathbf{F} = \sqrt{\frac{4\pi}{3}} \sum_{\nu=-1}^1 D_{\nu 0}^{(1)}(\hat{R}) r Y_1^\nu(\hat{\mathbf{r}}), \quad (\text{S26})$$

where $D_{\nu 0}^{(1)} = \sqrt{(1-\nu)!/(1+\nu)!} P_1^\nu(\cos \beta) e^{-i\nu\alpha}$ is the Wigner rotation-matrix element and P_1^ν the associated Legendre function. Evaluating the explicit forms of $D_{\nu 0}^{(1)}$ and Y_1^ν gives the compact expression

$$\hat{\mu} = \cos \alpha \sin \beta \hat{x} + \sin \alpha \sin \beta \hat{y} + \cos \beta \hat{z}. \quad (\text{S27})$$

The Euler angle γ does not appear for linear polarization. When $\beta = 0$ the two frames coincide and the laser polarization points along the MF z -axis, since the LF Z -axis is defined along the polarization direction (Fig. 1 of the main text); in this case α has no effect on the dipole integral either.

To identify the origin of the low-energy structures in the cross section, we decompose the angle-integrated z -polarized signal into its ℓ -resolved contributions, $\sum_m |d_{\ell m}(E)|^2$ (Fig. S1). The maxima at $E = 6.48$ and 12 eV are both controlled by the $\ell = 5$ sector. The first peak is not produced by the broad nonresonant $\ell = 1$ background alone; it appears when a localized $\ell = 5$ enhancement is superimposed on that smooth background. The second peak is even more directly tied to this low- ℓ sector, being dominated by $\ell = 5$ with only weak admixture from $\ell = 7$ and higher channels. This behaviour rules out an interpretation in terms of high- ℓ diffraction structure. Together with the independent channel-resolved phase evolution—large positive partial delays—this provides strong evidence that both features are shape resonances, i.e., transient trapping of the outgoing electron by the molecular cage potential before escape into the continuum.

An important angular property of the cross section deserves comment. At a polar emission angle $\vartheta = \pi/2$ with $\beta = 0$, and reciprocally at $\vartheta = 0$ when $\beta = \pi/2$, the cross section approaches zero—a feature *not* reflected in the time delay. This is illustrated in Figs. 2 and 4 of the main text and in Fig. S2. The origin is as follows. At $\beta = 0$ only the $\hat{z}$ -term of Eq. (S27) survives, so the dominant partial cross sections carry $m = 0$ [Fig. S2(a)]. Because the associated Legendre functions with $m = 0$ vanish at $\vartheta = \pi/2$, the

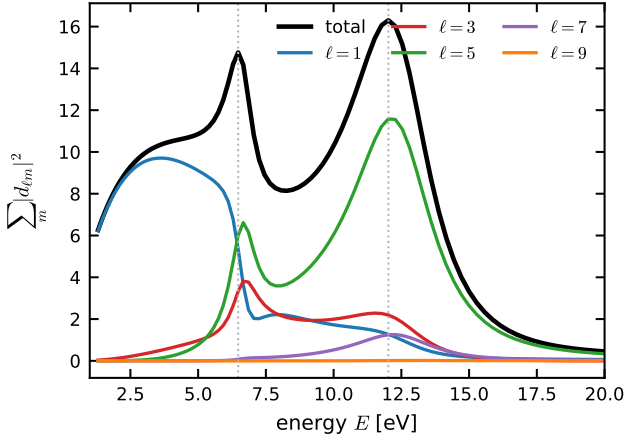


Figure S1: Partial-wave decomposition of the angle-integrated z -polarized dipole strength, $\sum_m |d_{\ell m}(E)|^2$, together with the total signal $\sum_{\ell m} |d_{\ell m}(E)|^2$, as a function of photoelectron kinetic energy E . The vertical dotted lines mark the first two maxima at $E = 6.48$ and 12 eV. Both structures require the $\ell = 5$ sector: the first peak arises from a localized $\ell = 5$ enhancement on top of a smooth $\ell = 1$ background, whereas the second peak is dominated by $\ell = 5$ with only minor higher- ℓ contributions.

cross section is suppressed in the direction perpendicular to the polarization. Conversely, at $\beta = \pi/2$ and $\vartheta = 0$ the nonzero partial matrix elements have $m \neq 0$ [Fig. S2(b)], but the corresponding associated Legendre functions vanish at $\vartheta = 0$, reproducing the same suppression. Neither situation affects the phase of the amplitude, so the time delay remains finite.

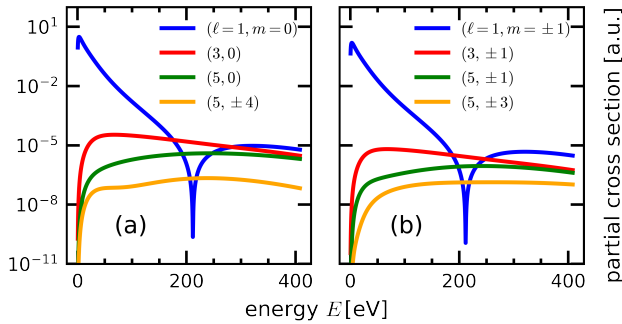


Figure S2: Partial cross section $\sigma_{\ell m}$ (logarithmic y-axis). (a) Contribution from the $\hat{z}$ -term of Eq. (S27). (b) Same for the $\hat{x}$ -term.

III. TRANSFORMATION FROM SPHERE TO CUBE

Following Ref. [1], the algebraic formulation for a three-dimensional *squircle* can be expressed as:

$$r^2 - \frac{4s^2}{L^2}(x^2y^2 + x^2z^2 + y^2z^2) + \frac{16s^4}{L^4}(x^2y^2z^2) = \frac{L^2}{4}, \quad (\text{S28})$$

where s denotes the adjustable squareness parameter. At $s = 0$, the geometry is a sphere with a diameter of L , while at $s = 1$, it shapes into a cube with an edge length of $\frac{L}{2}$. The effect of this geometric transformation on the photoionization of the diffracting target is illustrated in Fig. S3. To ensure that $s = 0$ corresponds to a bound state, we adjusted the potential V_0 to 2 a.u., resulting in a ground state energy $E_0^{s=0} = -0.83$ eV.

As depicted in Fig. S3, the resultant cross section and time delay distributions, computed in our method, evolve as the system's geometry transitions from a perfect sphere (leftmost panels) to a cube (rightmost panels), with s modulating the squareness. The two concentric circular fringes in images starting at $s=0$ adaptively evolve through square-like contours as s alters to produce a complex fringe pattern, a central hypocycloid with surrounding stacks of cusps, as s becomes unity. This vividly captures the evolution of the diffraction patterns in both the photoionization cross section and time delay from the spherical to cubical target symmetry.

IV. AVERAGING ANALYSIS

For a given emission direction $\hat{k}$ and molecular orientation (or, equivalently, polarization configuration) $\hat{R}$, we define the pair of quantities

$$I(E, \hat{k}, \hat{R}) = |\mathcal{D}^{(-)}(E, \hat{k}, \hat{R})|^2, \quad (\text{S29a})$$

$$N(E, \hat{k}, \hat{R}) = \text{Im}[\mathcal{D}^{(-)*} \partial_E \mathcal{D}^{(-)}], \quad (\text{S29b})$$

so that the differential cross section and time delay [Eqs. (S25) and (S22)] are

$$\sigma = \frac{4\pi^2\omega}{c} I, \quad \tau = \frac{N}{I}. \quad (\text{S30})$$

Because the prefactor $4\pi^2\omega/c$ is independent of angles, any angular average of the time delay must be weighted by the corresponding differential cross section, or equivalently by I . This is a direct consequence of the ratio structure of τ and ensures that directions contributing more signal carry more weight.

The emission-angle average at fixed orientation $\hat{R}$ is therefore

$$\bar{\tau}(E, \hat{R}) = \frac{\int N(E, \hat{k}, \hat{R}) d\Omega}{\int I(E, \hat{k}, \hat{R}) d\Omega}, \quad (\text{S31})$$

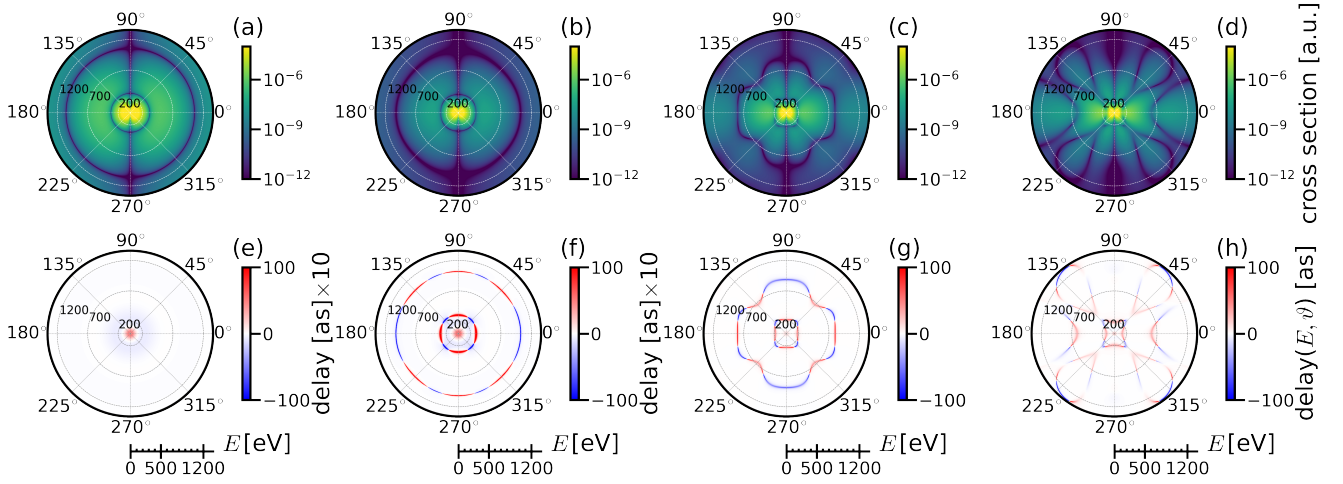


Figure S3: Systematic visualization of the evolution of photoionization cross-sections (top row) and time delays (bottom row) on the polar plane as a function of the degree-of-squareness parameter s , for an azimuthal angle $\varphi = 0$ and a polarization direction of $\beta = 0$. Each row consists of four figures, depicting a progressive transformation from a spherical geometry $s = 0$ (a,e) to a cubic geometry $s = 1$ (d,h), with intermediate values at $s = 0.5$ (b,f) and $s = 0.95$ (c,g).

with $d\Omega = \sin\vartheta d\vartheta d\varphi$, and the corresponding averaged cross section is $\bar{\sigma}(E, \hat{R}) = \frac{1}{4\pi} \int \sigma d\Omega$. Using the partial-wave expansion (S23) and the orthonormality of the spherical harmonics, these reduce to closed-form sums over partial waves,

$$\bar{\sigma} = \frac{4\pi^2\omega}{c} \frac{4\pi}{k^2} \sum_{\ell m} |d_{\ell m}|^2, \quad (\text{S32a})$$

$$\bar{\tau} = \frac{\text{Im} \left[\sum_{\ell m} d_{\ell m}^* \partial_E d_{\ell m} \right]}{\sum_{\ell m} |d_{\ell m}|^2}. \quad (\text{S32b})$$

At fixed polar angle ϑ the azimuthal average is defined analogously,

$$\tilde{\tau}(E, \vartheta, \hat{R}) = \frac{\int_0^{2\pi} N d\varphi}{\int_0^{2\pi} I d\varphi}, \quad (\text{S33})$$

with $\tilde{\sigma} = \frac{1}{2\pi} \int_0^{2\pi} \sigma d\varphi$. For the real spherical harmonics $Y_{\ell m}^R(\vartheta, \varphi) = \Theta_{\ell m}(\vartheta) \Phi_m(\varphi)$ used in the present work,

$$\frac{1}{2\pi} \int_0^{2\pi} \Phi_m \Phi_{m'} d\varphi = c_m \delta_{mm'}, \quad c_m = \begin{cases} 1, & m = 0, \\ 1/2, & m \neq 0. \end{cases} \quad (\text{S34})$$

Defining the azimuthal partial amplitude

$$A_m(E, \vartheta, \hat{R}) = \sum_{\ell} i^{-\ell} d_{\ell m}(E, \hat{R}) \Theta_{\ell m}(\vartheta), \quad (\text{S35})$$

the azimuthally averaged cross section and time delay be-

come

$$\bar{\sigma} = \frac{4\pi^2\omega}{c} \frac{16\pi^2}{k^2} \sum_m c_m |A_m|^2, \quad (\text{S36a})$$

$$\bar{\tau} = \frac{\sum_m c_m \text{Im}[A_m^* \partial_E A_m]}{\sum_m c_m |A_m|^2}. \quad (\text{S36b})$$

The azimuthal average is thus obtained analytically from the orthogonality of the Φ_m functions, without numerical quadrature.

Averages over the molecular orientation (or, equivalently, the polarization direction) $\hat{R} \equiv (\alpha, \beta, \gamma)$ are treated in the same cross-section-weighted manner,

$$\tau_{\text{avg}}(E, \hat{k}) = \frac{\int d\hat{R} N(E, \hat{k}, \hat{R})}{\int d\hat{R} I(E, \hat{k}, \hat{R})}, \quad (\text{S37})$$

with $d\hat{R} = \sin\beta d\beta d\alpha d\gamma$ and $\sigma_{\text{avg}} = \frac{1}{8\pi^2} \int d\hat{R} \sigma$. For an isotropic polarization average, the transition amplitude for polarization direction $\hat{F}$ is decomposed as $\mathcal{D}^{(-)}(\hat{k}; \hat{F}) = \sum_{i=x,y,z} F_i \mathcal{D}_i^{(-)}(\hat{k})$, and $\langle F_i F_j \rangle_{\hat{F}} = \delta_{ij}/3$ eliminates all cross terms. The polarization-averaged quantities are therefore

$$\sigma_{\text{pol}} = \frac{4\pi^2\omega}{c} \frac{1}{3} \sum_{i=x,y,z} |\mathcal{D}_i^{(-)}|^2, \quad (\text{S38a})$$

$$\tau_{\text{pol}} = \frac{\sum_i \text{Im}[\mathcal{D}_i^{(-)*} \partial_E \mathcal{D}_i^{(-)}]}{\sum_i |\mathcal{D}_i^{(-)}|^2}. \quad (\text{S38b})$$

Combining the azimuthal and isotropic polarization averages yields

$$\tilde{\sigma}_{\text{pol}} = \frac{4\pi^2\omega}{c} \frac{16\pi^2}{3k^2} \sum_m c_m \sum_i |A_m^{(i)}|^2, \quad (\text{S39a})$$

$$\tilde{\tau}_{\text{pol}} = \frac{\sum_m c_m \sum_i \text{Im}[A_m^{(i)*} \partial_E A_m^{(i)}]}{\sum_m c_m \sum_i |A_m^{(i)}|^2}. \quad (\text{S39b})$$

Finally, the fully averaged delay appropriate to an experiment insensitive to both emission direction and molecular orientation is

$$\langle \tau \rangle(E) = \frac{\int d\hat{R} \int d\Omega N}{\int d\hat{R} \int d\Omega I} = \frac{\int d\hat{R} \bar{\tau} \bar{\sigma}}{\int d\hat{R} \bar{\sigma}}, \quad (\text{S40})$$

with the total averaged cross section $\langle \sigma \rangle = \frac{1}{32\pi^3} \int d\hat{R} \int d\Omega \sigma$. Equations (S32)–(S39) are the formulas used in the present analysis. All reported time-delay averages are cross-section weighted, and the angular averages are evaluated analytically whenever possible.

V. NUMERICAL METHODOLOGY

The coupled radial equations (S15) are solved using the renormalized Numerov method [2], which propagates the ratio matrix $\mathbf{R}_n = \mathbf{Y}_{n+1} \mathbf{Y}_n^{-1}$ via the stable recurrence $\mathbf{R}_n = \mathbf{U}_n - \mathbf{R}_{n-1}^{-1}$, thereby avoiding the numerical instabilities associated with exponentially growing and decaying solution pairs. The same numerical framework is applied to both

bound- and continuum-state calculations for the model potential.

The radial grid extends from $r = 0$ to r_{max} with a uniform step size $\delta r = 0.01 a_0$. Because the asymptotic oscillation wavelength of the continuum wavefunction scales as $1/k$, the box size is adapted to the photoelectron wavenumber: we use $r_{\text{max}} = 350 a_0$ for $k \leq 1$ a.u. and $r_{\text{max}} = 100 a_0$ for $k > 1$ a.u., which in both regimes is sufficient for the potential to have decayed and for the asymptotic matching to the Riccati–Bessel functions [Eq. (S17)] to be reliable. Energy-normalized continuum states are computed on a uniform k grid with step $\delta k = 0.01$ a.u.

For the model potential (S18) used in the sphere-to-cube analysis (Sec. III), the single-center expansion of the continuum orbital is truncated at $\ell_{\text{max}} = 10$, giving $N_{\text{ch}} = (\ell_{\text{max}} + 1)^2 = 121$ coupled channels per energy.

For the *ab initio* static-exchange calculations on C_8F_8^- , three independent angular-momentum truncations enter the computation. The continuum photoelectron orbital [Eq. (S10a)] is expanded up to $\ell_{\text{max}} = 80$; the Poisson auxiliary functions [Eq. (S10b)] up to $\ell_{\text{max}} = 10$; and the single-center expansion of the occupied Hartree–Fock molecular orbitals [Eq. (S10c)] up to $\ell_{\text{max}} = 300$, which is required to reproduce the norm of the deepest-lying core orbitals to within 10^{-7} of unity. Outer-valence orbitals converge at considerably lower ℓ values; nevertheless, we adopt the same truncation for all orbitals to ensure a uniform treatment across the entire occupied manifold. For the single-center expansion of the occupied molecular orbitals, we employ Gauss–Legendre quadrature with 1000 nodes in each of the θ and ϕ directions.

The present three-dimensional methodology upgrades the approach earlier used for two-dimensional systems in Ref. [3].

-
- [1] M. Fernández Guasti and M. De La Cruz Heredia, *Diffraction Pattern of a Circle/square Aperture*, *J. Mod. Opt.* **40**, 1073–1080 (1993).
 [2] B. R. Johnson, *The renormalized Numerov method applied to*

calculating bound states of the coupled-channel Schrödinger equation, *J. Chem. Phys.* **69**, 4678–4688 (1978).

- [3] Sajad Azizi, *Three aspects of photo-ionization in ultrashort pulses*, Ph.D. thesis, Technische Universität Dresden (2023).